

Final Project

Evaluating Optimal Colorectal Cancer Screening Strategies in Taiwan: A Cost-Effectiveness Microsimulation Analysis

Introduction

Colorectal cancer (CRC) is the third most common cancer and the second leading cause of cancer-related death in Taiwan, particularly between 2008 and 2021 [1]. Notably, recent trends indicate a rising incidence of CRC among younger populations, with incidence rates increasing from 21 to 29.8 per 100,000 individuals aged 40–44 and from 40.4 to 49.8 per 100,000 among those aged 45–49 between 2010 and 2018 [2]. In response to this growing burden, periodic screening using the faecal immunochemical test (FIT) has been widely implemented and shown to be effective, reducing late-stage (Stage II or above) CRC incidence by 29% and overall CRC-related mortality by 35% [3]. As early-stage CRC is often asymptomatic, regular screening plays a critical role in the early detection of precancerous lesions and early-stage disease.

To address this burden, Taiwan's Health Promotion Administration introduced a biennial FIT-based screening programme in 2004 for individuals aged 50–69 years [4]. The programme was subsequently expanded to include those up to age 74 in 2016, and further extended in 2025 to individuals aged 45 [4,5]. In addition, individuals aged 40–44 years with a family history of CRC are eligible for free FIT screening [5]. These policy developments reflect ongoing efforts to enhance early detection and reduce both CRC incidence and mortality at the population level.

While FIT-based screening is well established as effective, recent evidence from Taiwan suggests that the timing of screening initiation may further influence outcomes. A large community-based study analysing 17 years of data from over 500,000 individuals found that those who began FIT screening between the ages of 40 and 49 experienced a 21–25% reduction in CRC incidence and a 34–39% reduction in CRC-related mortality compared with those who initiated screening at age 50 [6]. These findings indicate that earlier screening may provide additional benefits and highlight the importance of evaluating optimal starting ages and screening strategies for CRC.

Given the recent expansion of CRC screening eligibility and growing evidence supporting earlier screening, this study aims to evaluate the cost-effectiveness of different screening strategies in Taiwan. We focus on identifying the optimal starting age, stopping age, and screening frequency by comparing their effects on costs, quality-adjusted life years (QALYs), and CRC-related mortality reduction using a microsimulation model.

Method

I. Model Simulation Overview

We developed a CRC screening microsimulation model to simulate screening status, CRC incidence, CRC and all-cause mortality, CRC mortality reduction, QALYs, life expectancy, and direct healthcare system costs associated with FIT, follow-up colonoscopy, and CRC treatment over the lifetime of individuals aged 40–84 years in Taiwan (Fig. 1). The model evaluates the impact of alternative FIT screening strategies varying by screening initiation age, stopping age, and screening frequency, including strategies both within and outside the current government-funded screening eligibility range of 45–74 years.

The model progresses in yearly time steps, in which each individual ages by one year and may develop CRC, die from CRC or other causes, and accumulate costs and QALYs. Individuals are considered eligible for screening only if their current age falls within the strategy-specific screening age range and screening interval. Those outside the screening age range are not screened but continue to contribute to disease incidence, mortality, and health outcomes within the lifetime simulation framework.

The model adopts a stochastic Monte Carlo framework, in which transitions between health states, including healthy, CRC, and death, are determined probabilistically at each yearly cycle. Age-specific incidence and mortality rates reported per 100,000 population were converted into annual probabilities by dividing by 100,000 and applied at each time step. Screening, detection, and downstream clinical events were modelled using parameter values derived from published literature and national data sources. The detection effect parameter, representing the effect of screening on CRC mortality, was calibrated to reproduce the observed CRC mortality reduction associated with screening. Individual-level outcomes, including total healthcare costs, QALYs, life years, CRC incidence, and CRC mortality, were accumulated over lifetime and summarised across the simulated cohort to compare alternative screening strategies.

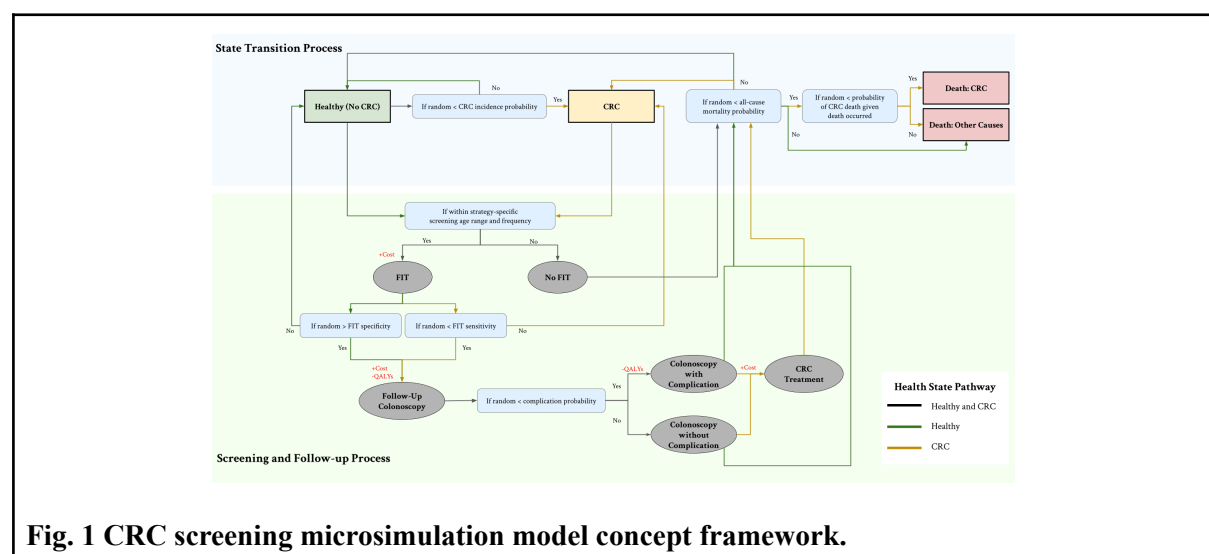


Fig. 1 CRC screening microsimulation model concept framework.

II. Model Parameter Inputs

Model input parameters included age-specific epidemiological inputs (CRC incidence, CRC mortality, and all-cause mortality), screening test characteristics (FIT sensitivity and specificity), cost parameters (FIT, follow-up colonoscopy, and CRC treatment), and utility values used to estimate QALYs (Table 1; Supplementary Table 1).

Table 1. Summary of model inputs.			
Category	Parameter	Value	Source
Epidemiological inputs	Age-specific CRC incidence rate	S2 Table	GBD 2021 (IHME) [7]
	Age-specific CRC mortality rate	S2 Table	GBD 2021 (IHME) [7]
	Age-specific all-cause mortality rate	S2 Table	GBD 2021 (IHME) [7]
Screening characteristics	FIT sensitivity	0.90	Medical literature [8]
	FIT specificity	0.83	Medical literature [8]
Costs (USD)	Cost of FIT	10	Healthcare provider websites [9]
	Cost of colonoscopy	150	Healthcare provider websites [10]
	Cost of CRC treatment	14,334	Medical literature [11]
Utilities/ QALY weights	Utility (healthy)	1.0	Assumed (perfect health)
	Utility (CRC)	0.7	Medical literature [12]
	Disutility (FIT false positive)	0.08	Medical literature [13]
	Disutility (colonoscopy discomfort)	0.0055	Medical literature [14]
	Disutility (colonoscopy complication)	0.0384	Medical literature [14]
Other parameters	Colonoscopy complication risk	0.015	Medical literature [8]
Calibrated parameter	Detection effect	0.84	Calibrated

III. Model Simulation Processes

Screening

All individuals start in a healthy state. Individuals receive FIT screening if their current age falls within the strategy-specific screening age range and screening interval, and the cost of screening is applied at each screening event.

For individuals with CRC, screening results in correct detection if a Monte Carlo random draw is below the FIT sensitivity. In such cases, individuals are marked as having detected cancer, and the cost of follow-up colonoscopy is applied. If a subsequent Monte Carlo random draw is below the probability of colonoscopy-related complications, a temporary reduction in QALYs is applied to reflect disutility (e.g., pain, anxiety, and adverse events such as bleeding or perforation) associated with complications.

For individuals without CRC, false-positive results may occur if a Monte Carlo random draw exceeds the FIT specificity. In these cases, individuals undergo unnecessary follow-up colonoscopy, incurring additional costs and a temporary reduction in QALYs associated with false-positive results, reflecting short-term psychological burden (e.g., anxiety). Similarly, if a subsequent Monte Carlo random draw is below the probability of colonoscopy-related complications, additional QALY loss associated with complications is applied.

CRC Incidence Rate

Individuals are marked as having CRC if a Monte Carlo random draw is below the age-specific incidence probability. This marks the start of the diagnosis event.

Mortality Rate

Individuals are marked as dead if a Monte Carlo random draw is below the age-specific all-cause mortality probability. Among individuals with CRC, the age-specific CRC mortality probability is used to determine whether death is attributable to CRC. If CRC is detected through screening, this probability is multiplied by a detection effect parameter to reflect the mortality reduction associated with screening. The conditional probability of dying from CRC given death is then calculated by dividing the CRC mortality probability by the all-cause mortality probability. Individuals are classified as dying from CRC if a Monte Carlo random draw is below this conditional probability.

One-Time Treatment Costs

A one-time cost for CRC treatment is added to individuals when they are first diagnosed with CRC. After the cost is applied, individuals are no longer considered newly diagnosed to ensure that the treatment cost is only incurred once.

QALYs and Life Expectancy

For individuals with CRC, a reduced utility value (0.7; Table 1) is added to total QALYs to reflect lower quality of life. For healthy individuals, a full utility value (1.0; Table 1) is added

to total QALYs to reflect perfect health. Life years are accumulated by increasing survival time by one year for each year the individual remains alive.

IV. Model Calibration

Model calibration was conducted by comparing the model-predicted CRC mortality reduction due to screening with an empirical target mortality reduction. The target mortality reduction was set to 35% based on real-world evidence reflecting the observed CRC mortality reduction among individuals aged 45–74 years undergoing FIT screening under the current Taiwan screening programme [3]. Calibration was performed using simulated annealing to identify the detection effect parameter that provided the best fit to this target.

To compute the predicted mortality reduction, the simulation was run under two scenarios: no screening and screening, each with a simulated cohort of 100,000 individuals. In the no-screening scenario, the detection effect was set to 1.0, meaning no mortality reduction was applied. Screening was disabled by setting the start and stop ages to 100, which is outside the simulated age range. In the screening scenario, the detection effect was varied during calibration to identify the value that best reproduced the target CRC mortality reduction, and screening was implemented from age 45 to 74 years with biennial FIT screening, reflecting the Taiwan screening strategy used as the reference scenario. For both scenarios, CRC deaths were counted, and the mortality reduction was calculated as:

$$1 - \frac{\text{CRC deaths under screening}}{\text{CRC deaths under no screening}}$$

The calibration error was calculated as the squared difference between the model-predicted mortality reduction and the empirical target mortality reduction. This is similar to the mean squared error approach, although only one calibration target was used.

Simulated annealing was used to search for the optimal detection effect. The initial detection effect of 0.7 was selected to approximate the expected CRC mortality reduction associated with FIT screening reported in Taiwan. The initial temperature was set to 1.0 to allow some worse candidate values to be accepted early in the search. At each iteration, a candidate detection effect was generated by randomly increasing or decreasing the current detection effect by up to 0.05. The candidate value was restricted to remain between 0.5 and 1.0.

If the candidate detection effect produced a lower error than the current value, it was accepted. If the candidate produced a higher error, it could still be accepted with a probability based on the simulated annealing acceptance rule computed as the exponential of the difference in error divided by the current temperature. This allows the calibration process to avoid getting stuck in a local optimum. The temperature was then multiplied by 0.98 at each iteration, gradually reducing randomness as calibration progressed. The detection effect with the lowest error was retained as the calibrated parameter. Based on this calibration process,

the detection effect used in the model is treated as a calibrated parameter rather than being directly derived from published data.

Additionally, to compare the effects of different screening strategies, mortality reduction was computed for alternative screening strategies using the calibrated detection effect. Each strategy was evaluated using a simulated cohort of 100,000 individuals. These strategies included start ages of 40, 45, and 50 years; stop ages of 69, 74, and 84 years; and annual or biennial screening frequencies. For each strategy, mortality reduction was calculated using the same comparison between the strategy-specific screening scenario and the no-screening scenario.

V. Software

The CRC screening microsimulation model was developed in Java (version 21.0.10) using Eclipse IDE (version 4.39.0), and downstream analyses, including statistical analysis and visualisations, were performed in R (version 4.5.2).

Results

The calibrated detection effect parameter was estimated to be approximately 0.86, producing a CRC mortality reduction of approximately 35%, which closely matched the empirical target mortality reduction used for calibration (Supplementary Table 3; Supplementary Fig. 1). These findings suggest that the model was successfully calibrated to reproduce the observed mortality benefit associated with FIT screening.

Across alternative screening strategies varying by screening initiation age, stopping age, and screening frequency, we identified strategies associated with the lowest healthcare system costs, highest QALYs, and greatest CRC mortality reduction (Table 2). The microsimulation model suggested that biennial screening beginning at age 50 years and stopping at age 69 years resulted in the lowest costs and highest QALYs (Tables 2–3; Fig. 2), whereas biennial screening beginning at age 40 years and stopping at age 84 years achieved the greatest CRC mortality reduction (Tables 2–3; Fig. 3). Compared with biennial screening beginning at age 40 years and stopping at age 84 years, biennial screening beginning at age 50 years and stopping at age 69 years was associated with an average reduction of approximately 956.32 USD in healthcare costs, an increase of 0.16 average QALYs, and a 0.18% lower average CRC mortality reduction.

Table 2. Best-performing CRC screening strategies.

Start Age (years)	Stop Age (years)	Frequency	Performance Metric	Average Cost (USD)	Average QALYs	Average CRC Mortality Reduction (%)
40	84	Biennial	Greatest CRC mortality reduction	1765.67	37.80	0.44
50	69	Biennial	Highest QALYs	809.35	37.96	0.26
50	69	Biennial	Lowest costs	809.35	37.96	0.26

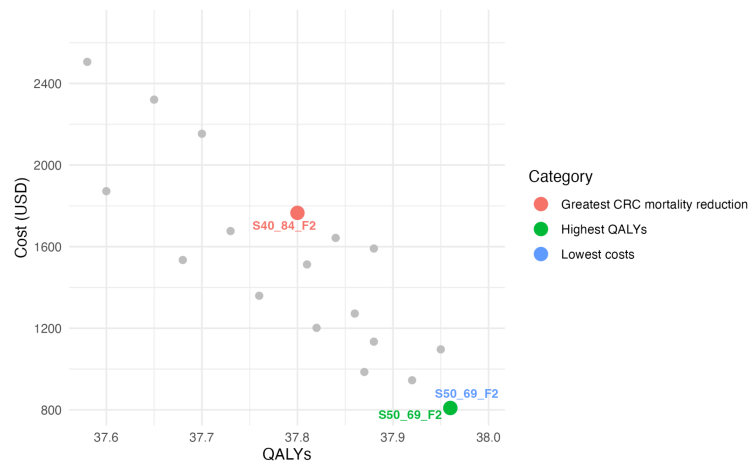


Fig. 2 Trade-offs between costs and QALYs across CRC screening strategies.

Table 3. Average outcomes across alternative CRC screening strategies.

Start Age (years)	Stop Age (years)	Frequency	Average Cost (USD)	Average QALYs	Average Mortality Reduction (%)
40	69	Annual	1534.57	37.68	0.18
40	69	Biennial	985.95	37.87	0.23
40	74	Annual	1872.13	37.60	0.33
40	74	Biennial	1272.10	37.86	0.29
40	84	Annual	2505.97	37.58	0.33
40	84	Biennial	1765.67	37.80	0.44
45	69	Annual	1359.46	37.76	0.21
45	69	Biennial	945.24	37.92	0.27
45	74	Annual	1676.63	37.73	0.25
45	74	Biennial	1134.20	37.88	0.35
45	84	Annual	2320.79	37.65	0.40
45	84	Biennial	1642.96	37.84	0.39
50	69	Annual	1202.08	37.82	0.25
50	69	Biennial	809.35	37.96	0.26
50	74	Annual	1512.91	37.81	0.34
50	74	Biennial	1096.60	37.95	0.31
50	84	Annual	2153.48	37.70	0.30
50	84	Biennial	1590.90	37.88	0.33

Highlighted rows represent best-performing screening strategies across selected outcome measures.

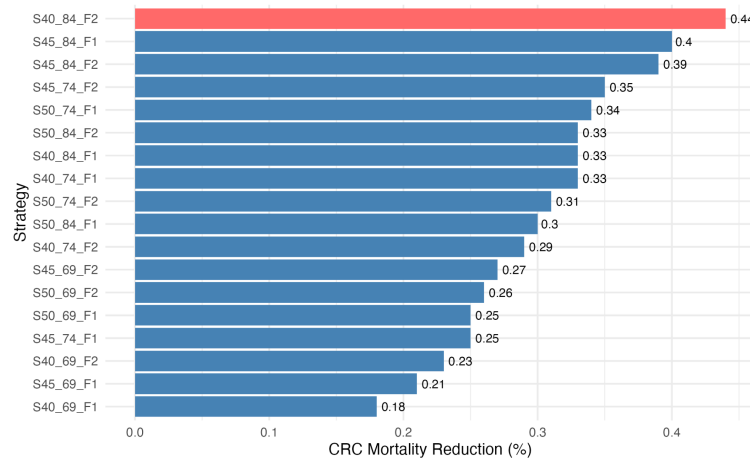


Fig. 3 Mortality reduction across CRC screening strategies.

Distinct trade-offs were observed between CRC mortality reduction, healthcare costs, and QALYs across alternative screening strategies. In comparing annual and biennial screening frequencies, no distinct pattern was observed for CRC mortality reduction alone. However, annual screening strategies were generally associated with substantially higher healthcare costs (Fig. 4) and comparatively lower QALYs (Fig. 5) compared with biennial screening strategies. In contrast, biennial screening strategies tended to cluster within the lower-cost (Fig. 4) and higher-QALY (Fig. 5) region of the cost-effectiveness space.

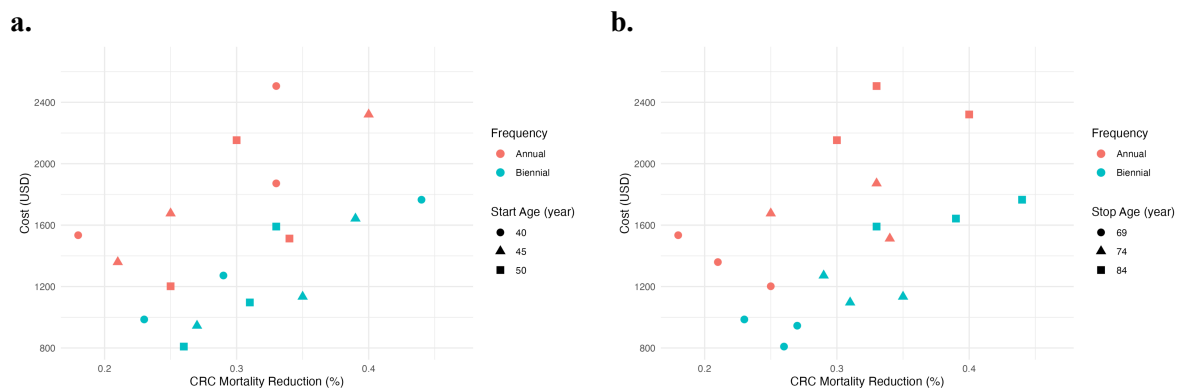
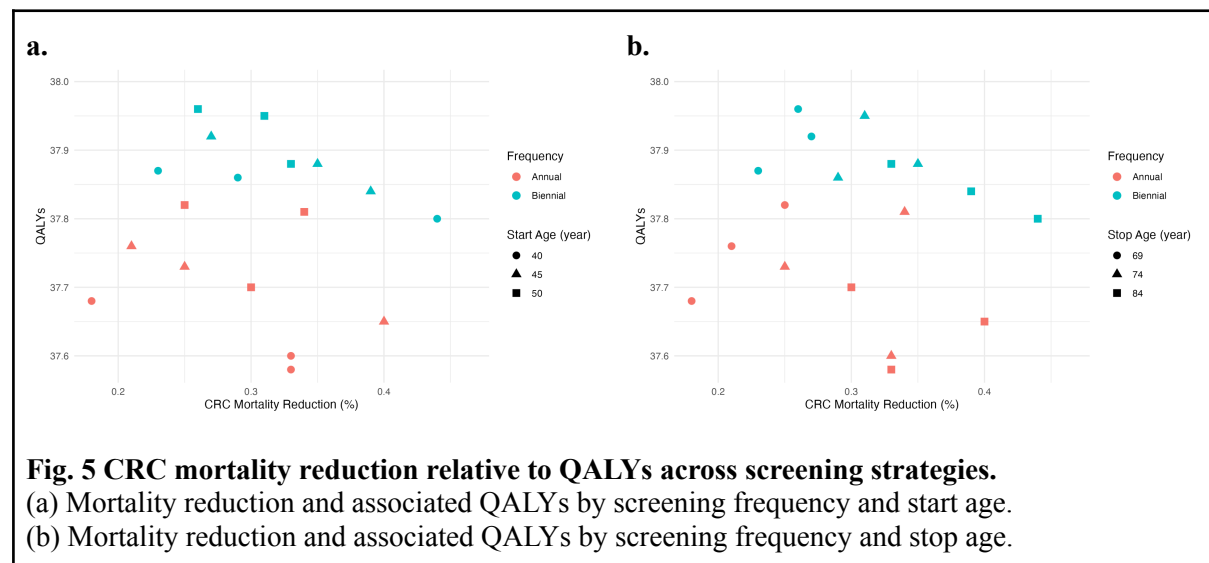


Fig. 4 CRC mortality reduction relative to healthcare costs across screening strategies.
(a) Mortality reduction and associated healthcare costs by screening frequency and start age.
(b) Mortality reduction and associated healthcare costs by screening frequency and stop age.



When comparing different screening start and stop ages, later stop ages were generally associated with greater CRC mortality reduction, with strategies stopping at age 84 years clustering towards the higher mortality reduction region, whereas strategies stopping at age 69 years clustered towards the lower mortality reduction region (Fig. 4b, 5b). In contrast, screening start age did not demonstrate a clear association with CRC mortality reduction (Fig. 4a, 5a). In terms of healthcare costs, after stratification by screening frequency, later stop ages were generally associated with higher costs across both annual and biennial screening strategies (Fig. 4). In terms of QALYs, later screening start ages and earlier stop ages were generally associated with comparatively higher QALYs across both screening frequencies (Fig. 5).

Discussion

This study aims to identify the optimal CRC screening strategy combination (i.e., screening start age, stop age, and screening frequency) in relation to QALYs, CRC mortality reduction, and direct healthcare system costs associated with FIT, follow-up colonoscopy, and CRC treatment, including screening age ranges and frequencies beyond those currently implemented in the Taiwan FIT screening programme (i.e., biennial screening for individuals aged 45–74 years). Our microsimulation model evaluated combinations of screening start ages of 40, 45, and 50 years; screening stop ages of 69, 74, and 84 years; and annual and biennial screening frequencies.

In terms of screening frequency, biennial screening strategies were generally associated with lower healthcare costs and higher QALYs compared with annual screening strategies, while no consistent pattern was observed for CRC mortality reduction alone. In terms of screening start and stop ages, screening beginning at age 50 years and stopping at age 69 years resulted in the lowest healthcare costs and highest QALYs, whereas screening beginning at age 40 years and stopping at age 84 years resulted in the greatest CRC mortality reduction. Notably, biennial screening from ages 50–69 years achieved an average CRC mortality reduction of

approximately 0.26, compared with approximately 0.44 for biennial screening from ages 40–84 years, highlighting the trade-off between maximising mortality reduction and optimising cost-effectiveness outcomes. Overall, these findings suggest that later screening start ages and earlier stop ages may be preferable when prioritising lower healthcare costs and higher QALYs, whereas earlier screening start ages and later stop ages may provide greater reductions in CRC mortality. These findings highlight the importance of carefully considering the screening age range when balancing cost-effectiveness and mortality benefits across CRC screening strategies.

However, screening was assumed to affect CRC mortality through early detection, but not CRC incidence, as the model does not explicitly incorporate adenoma development, detection, and removal within the disease transition process. Additionally, the model assumes that individuals diagnosed with CRC remain in the CRC state until death and does not explicitly model remission, cure, recurrence, transitions back to a disease-free state following treatment, or stage-specific disease progression. As CRC stage strongly influences survival, treatment costs, and QALYs, these assumptions may affect the estimation of long-term clinical and economic outcomes. Thus, future models should incorporate disease progression from healthy tissue to adenoma and CRC, together with stage-specific disease transitions and post-treatment outcomes.

Conclusion

This study demonstrated the impact of CRC screening age range on cost-effectiveness and mortality benefits, with biennial screening beginning at age 50 years and stopping at age 69 years yielding the most favourable cost-effectiveness outcomes, whereas biennial screening beginning at age 40 years and stopping at age 84 years achieved the greatest CRC mortality reduction. These findings highlight important trade-offs between healthcare costs, QALYs, and CRC mortality reduction when determining optimal screening strategies. Future work should incorporate adenoma progression, stage-specific CRC progression, and post-treatment disease transitions to better capture the natural history and long-term outcomes of CRC screening.

Appendix

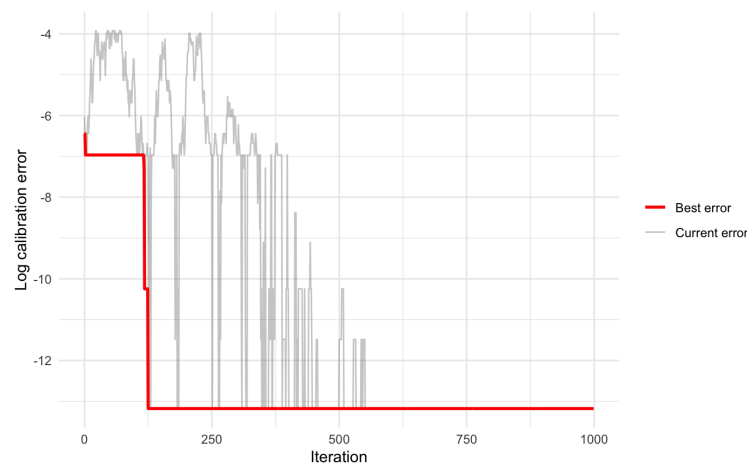
Supplementary Table 1. Model input parameters.		
Category	Parameter	Description
Epidemiological inputs	Age-specific CRC incidence rate	Age-specific CRC incidence per 100,000
	Age-specific CRC mortality rate	Age-specific CRC mortality per 100,000
	Age-specific all-cause mortality rate	Age-specific all-cause mortality per 100,000
Screening characteristics	FIT sensitivity	Probability of detecting CRC given disease
	FIT specificity	Probability of correctly identifying non-CRC
Costs (USD)	Cost of FIT	Cost per screening test
	Cost of colonoscopy	Cost of follow-up diagnostic procedure
	Cost of CRC treatment	One-time treatment cost
Utilities/ QALY weights	Utility (healthy)	Annual QALY weight for healthy state
	Utility (CRC)	Annual QALY weight while living with CRC
	Disutility (FIT false positive)	One-time QALY loss due to FIT false positive
	Disutility (colonoscopy discomfort)	One-time QALY loss per colonoscopy
	Disutility (colonoscopy complication)	One-time QALY loss per colonoscopy complication
Other parameters	Colonoscopy complication risk	Probability of complication
Calibrated parameter	Detection effect	Effect of screening on CRC mortality

Supplementary Table 2. Age-specific CRC incidence, mortality, and all-cause mortality rates in Taiwan, 2021. Rates are reported per 100,000 population.

Age Group (years)	CRC Incidence Rate (years)	CRC Mortality Rate (years)	All-Cause Mortality Rate (years)
40-44	35.79	9.15	182.65
45-49	60.32	15.44	285.30
50-54	108.23	26.85	431.36
55-59	155.81	41.09	590.75
60-64	212.46	61.08	827.85
65-69	260.08	84.49	1184.23
70-74	306.94	118.33	1769.17
75-79	454.23	210.84	3315.64
80-84	491.46	312.24	5651.01

Supplementary Table 3. Model calibration method and parameters.

Parameter	Value
Target CRC mortality reduction	0.35
Calibrated detection effect	0.836
Model-predicted CRC mortality reduction	0.349
Absolute calibration error	0.001
Optimisation method	Simulated annealing
Number of iterations	1000

Supplementary Fig 1. Calibration trace from the simulated annealing algorithm used to calibrate the detection effect parameter. The grey line represents the current calibration error across iterations, whereas the red line represents the lowest calibration error identified during optimisation. The calibration error decreased substantially during the early iterations followed by stabilisation, indicating successful convergence of the simulated annealing algorithm towards the calibrated detection effect parameter.

References

1. Cancer Statistics – Taiwan Cancer Registry. [cited 6 May 2026]. Available: https://twcr.tw/?page_id=1855&lang=en
2. 青壯年也要小心大腸癌！大腸癌公費篩檢擴大升級。In: 肝病防治學術基金會 [Internet]. [cited 6 May 2026]. Available: <https://www.liver.org.tw/journalView.php?cat=86&sid=1381&page=1>
3. Gastroenterological Society of Taiwan. The Taiwan guideline for colorectal cancer screening. Health Promotion Administration; Available: <https://www.dest.org.tw/DB/News/file/501-2.pdf>
4. 大腸癌_篩檢沿革。In: Taiwan Screening Website [Internet]. [cited 6 May 2026]. Available: <https://www.taiwanscreening.com/10>
5. Healthy Taiwan: Expanding Colorectal Cancer Screening for Better Protection. In: 衛生福利部國民健康署 [Internet]. [cited 8 May 2026]. Available: <https://www.hpa.gov.tw/EngPages/Detail.aspx?nodeid=1051&pid=18910>
6. Chiu H-M, Chen SL-S, Su C-W, Yen AM-F, Hsu W-F, Hsu C-Y, et al. Long-term effectiveness associated with fecal immunochemical testing for early-age screening. JAMA Oncol. 2025;11: 846–854. doi:[10.1001/jamaoncol.2025.1433](https://doi.org/10.1001/jamaoncol.2025.1433)
7. GBD Results. In: Institute for Health Metrics and Evaluation [Internet]. [cited 8 May 2026]. Available: <https://vizhub.healthdata.org/gbd-results>
8. Wong G, Hope RL, Howard K, Chapman JR, Castells A, Roger SD, et al. One-time fecal immunochemical screening for advanced colorectal neoplasia in patients with CKD (DETECT study). J Am Soc Nephrol. 2019;30: 1061–1072. doi:[10.1681/ASN.2018121232](https://doi.org/10.1681/ASN.2018121232)
9. 中華民國防癌篩檢中心醫療財團法人. [cited 8 May 2026]. Available: <http://www.tcdc.org.tw/largeIntestine.html>
10. 大腸鏡費用: 健保有全額給付大腸鏡檢查嗎? 傳統大腸鏡和無痛大腸鏡費用差別。In: 佳齡健康 [Internet]. [cited 5 May 2026]. Available: <https://www.healthspan.com.tw/b/colonoscopy-cost>
11. Chen P-C, Lee J-C, Wang J-D. Estimation of life-year loss and lifetime costs for different stages of colon adenocarcinoma in Taiwan. PLoS One. 2015;10: e0133755. doi:[10.1371/journal.pone.0133755](https://doi.org/10.1371/journal.pone.0133755)
12. Pourrahmat M-M, Kim A, Kansal AR, Hux M, Pushkarna D, Fazeli MS, et al. Health state utility values by cancer stage: a systematic literature review. Eur J Health Econ. 2021;22: 1275–1288. doi:[10.1007/s10198-021-01335-8](https://doi.org/10.1007/s10198-021-01335-8)
13. Matza LS, Howell TA, Fung ET, Janes SM, Seiden M, Hackshaw A, et al. Health state utilities associated with false-positive cancer screening results. Pharmacoeconom Open. 2024;8: 263–276. doi:[10.1007/s41669-023-00443-w](https://doi.org/10.1007/s41669-023-00443-w)
14. van Hees F, Zauber AG, Klabunde CN, Goede SL, Lansdorp-Vogelaar I, van Ballegooijen M. The appropriateness of more intensive colonoscopy screening than recommended in Medicare beneficiaries: a modeling study: A modeling study. JAMA Intern Med. 2014;174: 1568–1576. doi:[10.1001/jamainternmed.2014.3889](https://doi.org/10.1001/jamainternmed.2014.3889)